ISA/IEC 62443-3-3 / 4-2 • SL-3 TESTING CHEAT SHEET

Schneider Electric PowerLogic ION9000 Series Power Quality Meter · Component type: **Embedded Device (EDR) - power/energy meter** · Target SL = SL 3

Identity / scope	**ION9000 series** (METSEION92030 / 92830, 92040 with RD192 display, 92043 / 92044 Measurement Canada, 92130 / 92140 LVDC, 93030 / 93040 LVCS, 93130 / 93140 LVDC/LVCS, **ION9000T** 95030 / 95040 with high-speed transient capture). Same security model on all models; behaviour depends on firmware: **v3.0.0** replaced Telnet with SSH/SFTP, removed HTTP (HTTPS only) and moved Standard security to Legacy support; **v4.0.0** brought Secure ION; **v4.5.0+** cannot be downgraded. Read firmware on the display Nameplate screen first.
Where to work	**ION Setup** (Setup Assistant; Ctrl+click = Advanced mode), webpages `https://<meter>`, display (touch screen; Setup Menu), SFTP client (port 22, meter credentials). Security settings live in an encrypted Security Configuration File (.scf) edited and pushed with ION Setup (Security › Security Mode).
Ports (manual)	ION **7700**, Secure ION **7443**, Modbus TCP **502**, Modbus RTU over Ethernet 7701, DNP **20000**, DLMS **4059** (8 connections shared; 3 DNP, 1 DLMS), Modbus gateway 502 (32), EtherGate COM1 **7801** / COM4 **7802**, HTTPS **443** (10, HTTP redirect), SFTP **22** (2), IEC 61850 **102** (5), BACnet/IP 47808, SNMP 161, SMTP 25 out, NTP 123, DHCP 67/68; MQTT client to broker 8883. Two switched Ethernet ports, one IP; two RS-485 (COM1, COM4).
Defaults (manual)	Security mode: the manual says Advanced is default (Cybersecurity) and Legacy for new meters (Firmware) - read it from the report. **User1 / 0** (administrator), **User2 / 0** (viewer), User3-User50 inactive; display password **0**. HTTPS with HTTP redirect, SFTP, Modbus TCP, ION and Secure ION **enabled**; SSH terminal, SNMP (communities **public** / **private**), MQTT and BACnet/IP **disabled**. Webpages on; configuration by display **on**, by webpages off, Modbus access Read Only; broadcast time sync on. Valid login events not logged (priority 0). Self-signed certificate. Default IP 169.254.0.10 / 255.255.0.0. Factory access off. If user access information is lost the meter goes back to the factory and logged data is lost - passwords must be held in the site vault.
Advisories	**SEVD-2021-040-01**: CVE-2021-22701 (CSRF), -22702 (Telnet cleartext), -22703 (HTTP cleartext), before **V3.0.0**. **SEVD-2021-068-02**: CVE-2021-22714 (memory buffer, reboot or code execution, CVSS 9.8) before V3.0.0. **SEVD-2023-129-03** / CISA ICSA-23-229-03: CVE-2022-46680 (ION protocol cleartext, 8.8) before **4.0.0**; fix = Secure ION. Recheck SEVD and CISA at each assessment. No IEC 62443 certificate found.
How to read	Each row: the ION Setup, webpage or display path, the read-only check, then the V&V evidence. Login, lockout and timeout tests only on a lab or spare meter with written authorization. The meter feeds revenue and power quality data; resets, templates, firmware and test mode change or clear it.

Project / config capture & evidence handling - capture the device backup + config export

Do this BEFORE working the checklist - screenshots and walking the config screens are not enough on their own. Capture the device's own backup / project export so nothing is missed:
1. Pull the native backup / project + full config export first (read-only, before anything): **ION Setup records and screenshots** through the `<ip>/TLS:7443` override (no fallback to plain ION): the site administrator's ION Setup Setup Assistant › Reports › Meter Configuration › Display › Save As .txt (security settings, user privileges); Reports › Event Log › Display (All available records) › Save As CSV; Communications › Advanced Ethernet › Protocols, SNMP, SysLog and NTP tabs, Serial Settings COM1 and COM4, Clock › Timezone and the Configuration Checklist; Template › Save to PC only if the site approves. Webpages: Setup (Ethernet with Advanced Settings, Serial Port, SNMP Parameters, NTP, SMTP, Date & Time, HTTPS) and Diagnostics › Statistics › Meter - Basic. SFTP listing (only) of the documents, IEC61850 and web folders. Photos: display Nameplate (model, serial, feature set, firmware, CRC), revenue lock icon (LED only if visible without opening the sealed cover), seals, terminal, USB and optical port covers. Note that the site's .scf files and SECURITY.LOG exist (do not copy them). **Never** open the Security wizard (Security Mode › Edit), press TimeSync or DST Settings, Template Send or Upgrade, Edit or Send in Register Edit, Edit on Logging › Memory or Advanced-mode setup registers, any reset, Test Mode or Meter Restart, webpage Apply, certificate Upload, Generate or Delete, Ethernet Port Enable, the Advanced Security Password Updater, factory access, the revenue lock switch, or SFTP upload or delete.
2. Export the security-relevant settings: user accounts / roles / password policy, comms / port / protocol enable-disable, and any certificate / key / CIP-Security settings - as files and screenshots.
3. Collect firmware & logs: firmware / boot revision, and the event / audit / diagnostic logs and fault / alarm buffers (exports) where the device provides them.
4. Capture the device identity: nameplate (catalog + serial), firmware / hardware rev, installed modules / option cards, and slot / network address - **photograph** the device, faceplate / HIM and panel labels.
5. Name + protect (keep everything): `<site>_<zone>_<assetTag>_<vendor>_<model>_<fwRev>_<YYYYMMDD-HHMM>_<initials>` with the native project / backup extension + the exports; SHA-256 hash the backup; store **only** on the encrypted evidence share; the backup may hold credentials / keys - **never** email it in cleartext and follow the project retention / classification policy. Do not save device passwords in tool profiles.
6. Then analyze offsite against the product-specific checklist, the product manuals, vendor research and experience; fill in the tool and auto-generate the report - and remember the manual fields the tool can't auto-write: the **executive-summary narrative**, the **overall assurance level + justification**, and the **punch-list**.

FR 1 — Identification & Authentication Control (IAC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 1.1	Human user identification & authentication	Advanced security: per-user login (User1-User50) for ION Setup, webpages and SFTP; the display asks for the display password. Legacy security support (standard.scf) is one numeric password for User1 (0-99999999, 8 digits). The manual is inconsistent on which mode new meters ship in - confirm the mode in the Reports › Meter Configuration. **Test:** open the webpage in a new browser -> login prompt before data. **V&V evidence:** Meter Configuration Report; login page screenshot.	Advanced security; every interface authenticates.	Legacy mode = Partial.
SR/CR 1.2	Software process & device identification & authentication	ION and Secure ION carry user credentials, then a session token. Modbus, DNP, DLMS, IEC 61850 and BACnet/IP are unencrypted and not described as authenticated (manual risk table). Modbus Access defaults to Read Only. MQTT checks the broker against CAfile.pem, optional username and password. **V&V evidence:** Security Options and Protocols screenshots; zone drawing.	Unauthenticated protocols disabled or confined; Modbus Read Only or None.	Partial - compensating.
SR/CR 1.3	Account management	Up to 50 users; User3-User50 inactive until set; users added or removed in the Security wizard; credentials do not expire. Confirm each active user is a named person or system, User2 is changed or unused, and leavers are removed. **V&V evidence:** user list from the report; site account register.	Active accounts named, needed and reviewed.	
SR/CR 1.4	Identifier management	User names are fixed slots (User1-User50); renaming is not described. Tie each slot to a person or system by register. **V&V evidence:** register vs report.	Every slot traceable to an owner.	Partial - procedural.
SR/CR 1.5	Authenticator management	Confirm defaults changed: User1 and User2 password **0**, display password **0**, SNMP public / private. The Configuration Checklist flags a default password in use. Password Updater exports (.txt) and SECURITY.LOG are plain text; .scf files sit in C:\ProgramData\Schneider Electric\ION Setup\Security - all handled as secrets. Ask the site; do not try defaults on a live meter. **V&V evidence:** Configuration Checklist screenshot; administrator statement.	No default passwords or communities; credential files protected.	
SR 1.6	Wireless access management	No wireless interface. Record as not applicable. **V&V evidence:** photo of ports.	N/A - no wireless.	N/A - justification.
SR/CR 1.7	Strength of password-based authentication	Set in the Security wizard (read from the report or an administrator statement; do not open the wizard): Password Complexity (IEEE 1686-2013 or NERC CIP-007-5) and Minimum Password Length (up to 16; 6 or more with complexity). Passwords up to 16 characters, ASCII 32-126; no ageing or history (manual: follow site policy). Some ION clients accept only 6 alphanumeric characters - check the site software. **V&V evidence:** report, or administrator statement if not shown.	Complexity on; length meets site SL-3 policy.	Partial - no ageing.
SR/CR 1.8	Public key infrastructure (PKI) certificates	Self-signed certificate is used for HTTPS and Secure ION. Setup › HTTPS: upload a CA-signed .pem, generate a new self-signed one or delete the custom one. ION Setup View Certificate Info shows the SHA1 thumbprint. Record issuer and validity. **V&V evidence:** HTTPS page and certificate screenshots. **Ext. tools:** `nmap -Pn -p 443,7443 --script ssl-cert <meter>`	CA-signed certificate in use, or self-signed trusted by procedure.	
SR/CR 1.9	Strength of public key-based authentication	Client certificate authentication is not described. MQTT validates the broker certificate chain against CAfile.pem. **V&V evidence:** note; MQTT settings if used.	Gap recorded; compensated by network controls.	Partial.
SR/CR 1.10	Authenticator feedback	Check password fields are masked and failed logins give no hint (not described in the manual). **Test:** (lab) wrong password -> generic failure. **V&V evidence:** screenshot.	Masked entry; generic failure.	
SR/CR 1.11	Unsuccessful login attempts	Set in the Security wizard › Select protocol lockout options (read from the report, do not open the wizard): lockout after unsuccessful attempts per protocol and lockout duration. Ethernet failures count for ION and Secure ION together; serial is separate; a locked user can still log in over another protocol. The display locks after too many password attempts. **Test:** (lab only) exceed the limit -> lockout event logged. **V&V evidence:** lockout settings (report or administrator statement); event log entry.	Lockout set for every protocol in use.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 1.12	System use notification	No login banner is described. Post the notice on the access path (jump host, ION Setup workstation). **V&V evidence:** note.	Notice shown by the access path, or gap recorded.	Not supported.
SR 1.13	Access via untrusted networks	Manual: no internet access; encrypt external links with a tunnel, TLS wrapper or Secure ION. Confirm no meter port is reachable from office or internet networks. **V&V evidence:** firewall rules; scan from the untrusted side. **Ext. tools:** `nmap -Pn -sT -p 22,23,80,102,443,502,4059,7443,7700,7701,7801,7802,20000 <meter>` from the untrusted side (authorized)	Meter reachable only from the metering zone.	
CR 1.14	Strength of symmetric key-based authentication	No symmetric-key authentication is described. Record as not used. **V&V evidence:** note.	N/A - not used.	N/A - justification.

FR 2 — Use Control (UC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 2.1	Authorization enforcement	Per-user permissions: Read, Time Sync, Peak Demand Reset, Comm Config, Test Mode, Full Meter Config, Security Config. Security Options: Enable Web, Allow setup by webpages (default off), Allow Front Panel programming (default on), Modbus Access None / Read Only / Read/Write, Allow Broadcasting Time Synchronization. Revenue lock blocks revenue settings, resets and upgrades. **Test:** (lab) log in as a viewer -> Setup changes refused. **V&V evidence:** permission matrix from the report.	Least privilege per user; write paths limited to need.	
SR/CR 2.2	Wireless use control	No wireless interface. N/A. **V&V evidence:** none required.	N/A.	N/A - justification.
SR 2.3	Use control for portable & mobile devices	The RD192 display has a USB host (Type A) and device (Mini-B) port and an ANSI optical port; their use is not described in the user manual - confirm covers fitted and sealed. ION Setup laptops must be site-managed. **V&V evidence:** port cover photos; laptop register.	Ports covered; only managed laptops connect.	Ask Schneider for USB use.
SR/EDR 2.4	Mobile code	Webpages and scripts are served from the meter web folder; SFTP users have full control of web and documents folders, so custom pages can be added. List the web folder and compare with the firmware package. **V&V evidence:** SFTP listing (read only).	Only approved web content on the meter.	EDR-specific.
SR/CR 2.5	Session lock	Inactivity timeout requires sign-in again for webpages, display and Secure ION; ION session tokens expire after 5 minutes idle; the display asks for its password again after the password timeout. **Test:** (lab) stay idle -> login required. **V&V evidence:** lab result.	Idle sessions end.	
SR/CR 2.6	Remote session termination	No administrator session list is described; sessions end by timeout. Supervisors can delete user accounts. **V&V evidence:** note.	Sessions end by timeout; accounts revocable.	Partial.
SR/CR 2.7	Concurrent session control	Manual limits: 8 shared ION / Secure ION / Modbus / DNP / DLMS connections (3 DNP, 1 DLMS), HTTPS 10, SFTP 2, EtherGate 1 per port, IEC 61850 5, Modbus gateway 32. **V&V evidence:** protocol settings.	Limits known; unused protocols off.	
SR/CR 2.8	Auditable events	Event log records configuration changes (with user), energy system events, invalid logins (priority 128) and lockouts (255). **Valid logins are priority 0 - not logged by default**; events at or below cutoff 5 are not logged or sent to syslog. Read the values; if valid logins are 0-5 recommend the site raises them (owner action). **V&V evidence:** Event Priorities values (report or administrator statement); event log CSV.	Logins, failures, lockouts and changes logged.	
SR/CR 2.9	Audit storage capacity	Event log holds 5,000 records by default, then overwrites. Designated Reader (EventLogCtl module) marks entries unread until reviewed; syslog keeps copies. **V&V evidence:** Designated Reader setting (Advanced mode, view only); syslog settings.	Logs reviewed or offloaded before overwrite.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 2.10	Response to audit processing failures	Designated Reader marks entries unread; notifying the reader needs a customized framework (manual). An event is logged if syslog is enabled with no server IP. No other failure response is described. **V&V evidence:** Designated Reader and any notification framework; syslog status.	Unread-log review or syslog in place.	Partial.
SR/CR 2.11	Timestamps	Clock › Timezone: Time Sync Source (COM1, COM4, ETHERNET, ETHERNET - ION, - DNP, - PTP, IRIG-B), Clock Source, Time Sync Type (UTC for NTP / PTP); NTP tab: NTP / SNTP, server, event logging threshold. Time source changes are logged. Disable broadcast time sync where not needed. **V&V evidence:** Timezone and NTP screenshots.	One trusted time source; changes logged.	
SR/CR 2.12	Non-repudiation	The meter embeds the user when a change is made; per-user accounts make actions attributable. Valid logins need a priority above 5 to be recorded. **V&V evidence:** event log entries with user names.	Changes attributable to a named user.	
EDR 2.13	Use of physical diagnostic & test interfaces	Display setup (front panel programming on by default, password 0), USB and optical ports, RS-485 COM1 / COM4, revenue lock switch, test mode. Factory access (Schneider diagnostic code over a secure protocol) is off by default, time limited and needs user privilege - confirm it is off. **V&V evidence:** Security Options; factory access state; photos.	Front panel programming off or passworded; factory access off.	

FR 3 — System Integrity (SI)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 3.1	Communication integrity	Protected: HTTPS, SFTP, Secure ION, Secure EtherGate, MQTT over TLS. Unprotected: ION, Modbus, DNP, DLMS, IEC 61850, BACnet/IP. CVE-2022-46680 (ION cleartext) fixed by Secure ION in 4.0.0. Confirm clients use Secure ION and plain ION is off. **V&V evidence:** Protocols tab; client connection settings; scan. **Ext. tools:** `nmap -Pn -sT -p 22,23,80,102,443,502,4059,7443,7700,7701,7801,7802,20000 <meter>`; `nmap -Pn -sU -p 161,47808 <meter>`	Configuration over Secure ION or HTTPS only; cleartext confined.	Partial - compensating.
SR/EDR 3.2	Protection from malicious code	Firmware is digitally signed and only Schneider-signed firmware is accepted. The SFTP documents folder accepts any file type. No other malware protection on the meter. **V&V evidence:** firmware version; SFTP listing.	Signed firmware; no unexpected files.	EDR-specific.
SR/CR 3.3	Security functionality verification	Meter Configuration Report and Configuration Checklist verify settings. Security configuration survives upgrades - review it after each one. **V&V evidence:** report dated after last change.	Settings verified after changes.	
SR/CR 3.4	Software & information integrity	Record firmware and firmware CRC (display Nameplate) and feature set (also webpage Diagnostics › Statistics › Meter - Basic). Revenue meters keep an upgrade log (retrieval via Schneider support). **V&V evidence:** Nameplate photo.	Firmware matches the approved signed release.	
SR/CR 3.5	Input validation	CVE-2021-22714 (memory buffer) fixed in V3.0.0. No robustness tests on live meters; bench only. **V&V evidence:** firmware version vs advisory.	Fixed firmware; no live tests.	
SR/CR 3.6	Deterministic output	Digital and analog outputs may change state during resets, option module resets, firmware upgrades and option module firmware upgrades; the manual says not to use the meter for critical control. Record what outputs drive and any IEC 61850 control. **V&V evidence:** I/O configuration screenshots.	Outputs not used for protection; behaviour documented.	
SR/CR 3.7	Error handling	Not described. Check webpage errors reveal no secrets. **V&V evidence:** note.	No sensitive data in errors.	Not documented.
SR/CR 3.8	Session integrity	ION sessions use a non-transferable token after the first authenticated request; webpages are HTTPS only. Cookie and web session handling is not described - check the response headers. **V&V evidence:** header output. **Ext. tools:** `curl -skI https://<meter>/`	Sessions protected by TLS and tokens.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 3.9	Protection of audit information	Master Reset clears event, data and waveform logs; revenue-locked meters block revenue resets. The permission needed for Master Reset is not stated - confirm in a lab or with Schneider; make sure syslog keeps a copy. **V&V evidence:** lab result or Schneider statement; syslog receipt.	Log clearing limited; copies kept off the meter.	
EDR 3.10	Support for updates	Upgrade with the latest ION Setup (Template › Upgrade, ZIP package); pre-2.1.0 meters go to 2.1.0 first; 4.5.0+ cannot be downgraded; revenue-locked meters cannot be upgraded; upgrades clear recorded data and overwrite webpages and templates. Confirm current firmware and a patch plan. **V&V evidence:** version vs latest on se.com.	Supported, current firmware; process defined.	EDR-specific.
EDR 3.11	Physical tamper resistance & detection	Revenue lock switch under a sealable cover with LED and display icon; anti-tamper sealing points; sealable terminal covers; Measurement Canada sealed model 92044. **V&V evidence:** seal and LED photos.	Locked and sealed where required; tampering evident.	EDR-specific.
EDR 3.12	Provisioning product supplier roots of trust	Signed firmware implies a Schneider verification key; key handling is not described - ask Schneider. **V&V evidence:** Schneider statement.	Supplier root of trust present.	Ask Schneider.
EDR 3.13	Provisioning asset owner roots of trust	Owner CA-signed certificate (.pem) via Setup › HTTPS for HTTPS and Secure ION; owner CA file CAfile.pem for MQTT. **V&V evidence:** certificate details.	Owner certificate provisioned.	
EDR 3.14	Integrity of the boot process	Not described. Ask Schneider. **V&V evidence:** Schneider statement.	Boot integrity present, or gap recorded.	Ask Schneider.

FR 4 — Data Confidentiality (DC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 4.1	Information confidentiality	Settings are kept in an encrypted .scf; SECURITY.LOG and Password Updater exports are plain text on the ION Setup PC. MQTT password is write-only and deleted on upgrade or template load. Cleartext protocols as in 3.1. **V&V evidence:** file handling note; protocol settings.	Credential files protected; cleartext confined.	Partial.
SR/CR 4.2	Information persistence	Wipe in order: overwrite template, delete SFTP documents, custom certificate, webpages and CID file, restore default security, overwrite Factory registers, Master Reset, overwrite Comm registers. Physical destruction recommended. Lost credentials mean return to factory with data loss. **V&V evidence:** disposal record.	Wipe or destruction recorded.	
SR/CR 4.3	Use of cryptography	TLS versions and ciphers are not listed in the manual - scan HTTPS and Secure ION. **V&V evidence:** TLS scan output. **Ext. tools:** `nmap -Pn -p 443,7443 --script ssl-enum-ciphers <meter>`	Current TLS only.	

FR 5 — Restricted Data Flow (RDF)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 5.1	Network segmentation	Two switched Ethernet ports share one IP and must not join two networks; RSTP for rings. EtherGate and Modbus gateway bridge Ethernet to RS-485 devices - include them in the zone. **V&V evidence:** zone drawing; gateway settings.	Meter, daisy chain and serial devices in one zone.	
SR 5.2	Zone boundary protection	Boundary firewall allows only the energy system and engineering hosts, preferably to 7443 and 443; change default ports per the manual if site policy wants it. **V&V evidence:** firewall rules.	Boundary filters meter traffic.	N/A in device.

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR 5.3	General purpose person-to-person communication restrictions	SMTP (outgoing 25) sends alarm or log e-mail. Confirm it is off or uses an internal relay only. **V&V evidence:** SMTP page screenshot.	No uncontrolled e-mail from the meter.	
SR 5.4	Application partitioning	Single-purpose meter. Record as not applicable. **V&V evidence:** none required.	N/A.	N/A - justification.

FR 6 — Timely Response to Events (TRE)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 6.1	Audit log accessibility	Event log via ION Setup Reports (CSV or TXT; range options All available records or Last), display Events screen and syslog (UDP, server set by IP address only). **V&V evidence:** exported event log; syslog sample.	Authorized staff can read logs off the meter.	
SR/CR 6.2	Continuous monitoring	Syslog maps priorities to severities (High = 2, Medium = 4, Low = 5); SNMP v2 traps (four receivers, AuthenticationFailure); alarms and e-mail. Forward to the SIEM. **V&V evidence:** syslog and SNMP trap settings.	Security events reach monitoring.	

FR 7 — Resource Availability (RA)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 7.1	Denial of service protection	Connection limits per protocol; CVE-2021-22714 could reboot the meter (fixed V3.0.0). No DoS tests on live meters. **V&V evidence:** firmware version.	Fixed firmware; boundary rate limits.	
SR/CR 7.2	Resource management	Logging › Memory (Status tab) shows each log's interval, depth and share of memory and the total in use; logging faster than 60 s risks data loss on power failure. View only - editing log depth clears that recorder. **V&V evidence:** Memory screen.	Log load sized to the meter.	
SR/CR 7.3	Control system backup	Keep the site .scf (encrypted), a Template › Save to PC copy, custom webpages (SFTP copy of web) and IEC 61850 CID file. **V&V evidence:** backup list with dates.	Current configuration backups held securely.	
SR/CR 7.4	Control system recovery & reconstitution	Recovery = firmware, template, .scf Send and webpages from backup; the site records energy values first (Register Edit, view only) because upgrades clear data. **V&V evidence:** recovery procedure.	Recovery procedure documented and tested off-line.	
SR 7.5	Emergency power	Control power ride-through typical 100 ms at 120 V AC, 400 ms at 240 V AC, 1200 ms at 480 V AC; use a UPS for heavy logging (manual). **V&V evidence:** power design.	Supply resilient where data is critical.	Infrastructure.
SR/CR 7.6	Network & security configuration settings	Compare the Meter Configuration Report and protocol, SNMP, syslog, NTP, serial and MQTT settings with the hardened baseline. **V&V evidence:** report vs baseline.	Settings match the baseline.	
SR/CR 7.7	Least functionality	Disable what is not used: plain ION, Modbus RTU over Ethernet, DNP, DLMS, BACnet, EtherGate, SNMP, MQTT, SMTP, DPWS where possible, unused RS-485 ports (Protocol = None), webpages if not needed; keep Secure ION. **V&V evidence:** Protocols tab; scan. **Ext. tools:** `nmap -Pn -sT -p 22,23,80,102,443,502,4059,7443,7700,7701,7801,7802,20000 <meter>`; `nmap -Pn -sU -p 161,47808 <meter>`	Only required services enabled.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / firmware
SR/CR 7.8	Control system component inventory	Record commercial reference (Factory module Options register, on meters shipped with 4.6.0 or later), serial, feature set, firmware, CRC, MAC, IP, option modules and display firmware. **V&V evidence:** inventory sheet.	Complete inventory.	

External Test Tools — how to use

Tool / method	Verifies	How to use — steps / example	Constraints
ION Setup	Report, event log, settings (most rows)	ION Setup Setup Assistant › Reports › Meter Configuration / Event Log › Save As.	No Finish, Send, Upgrade, resets.
Browser	Webpages (1.8, 3.8, 5.3)	`https://<meter>`; Setup pages view only.	No Apply, certificate actions.
SFTP client	Folder listing (2.4, 3.2)	`sftp -P 22 <user>@<meter>` then `ls` only.	No put, rm.
nmap	Exposure (1.13, 3.1, 7.7)	`nmap -Pn -sT -p 22,23,80,102,443,502,4059,7443,7700,7701,7801,7802,20000 <meter>`; `nmap -Pn -sU -p 161,47808 <meter>`	Authorized, low rate.
nmap ssl / curl	Certificates, TLS (1.8, 3.8, 4.3)	`nmap -p 443,7443 --script ssl-cert,ssl-enum-ciphers <meter>`	Read-only.
Schneider documents	Grounding (all)	7EN02-0390-10; SEVD-2021-040-01; SEVD-2021-068-02; SEVD-2023-129-03.	-

Controlled - Project Use · Independent Assurance Services · ISA/IEC 62443-3-3 / 4-2 · Product-specific companion to the 62443 assessment tool; confirm exact menu paths / CLI against the deployed firmware and the project hardening baseline. ISOLATED tools run only in a representative/lab environment with written authorization.